PYTHON ACTIVITY-BASED LEARNING (ABL)

CASE STUDY 2: ATTENDANCE RECOVERY COMPUTATION

1. Problem Statement

Write a Python program to calculate a student's current attendance percentage and find the minimum number of consecutive classes the student must attend to reach the 75% attendance requirement.

2. Objectives

1. Calculate attendance percentage using attended and total classes.

2. Understand variables, arithmetic operators, and if-else conditions.

3. Find the minimum future classes needed to reach 75% attendance.

4. Display the result clearly.

3. Assumptions

• Minimum required attendance is 75%.

• Attended classes (A) and total classes (T) are whole numbers.

• Total classes must be greater than zero, and attended classes cannot exceed total classes.

• The student attends every future class and does not miss any.

• These are assignment assumptions; actual college attendance rules may differ.

4. Algorithm

1. Start.

2. Input attended classes (A) and total classes (T).

3. If T is zero, A is negative, or A is greater than T, display invalid input.

4. Calculate percentage P = (A / T) × 100.

5. If P is at least 75, display that the attendance requirement is satisfied.

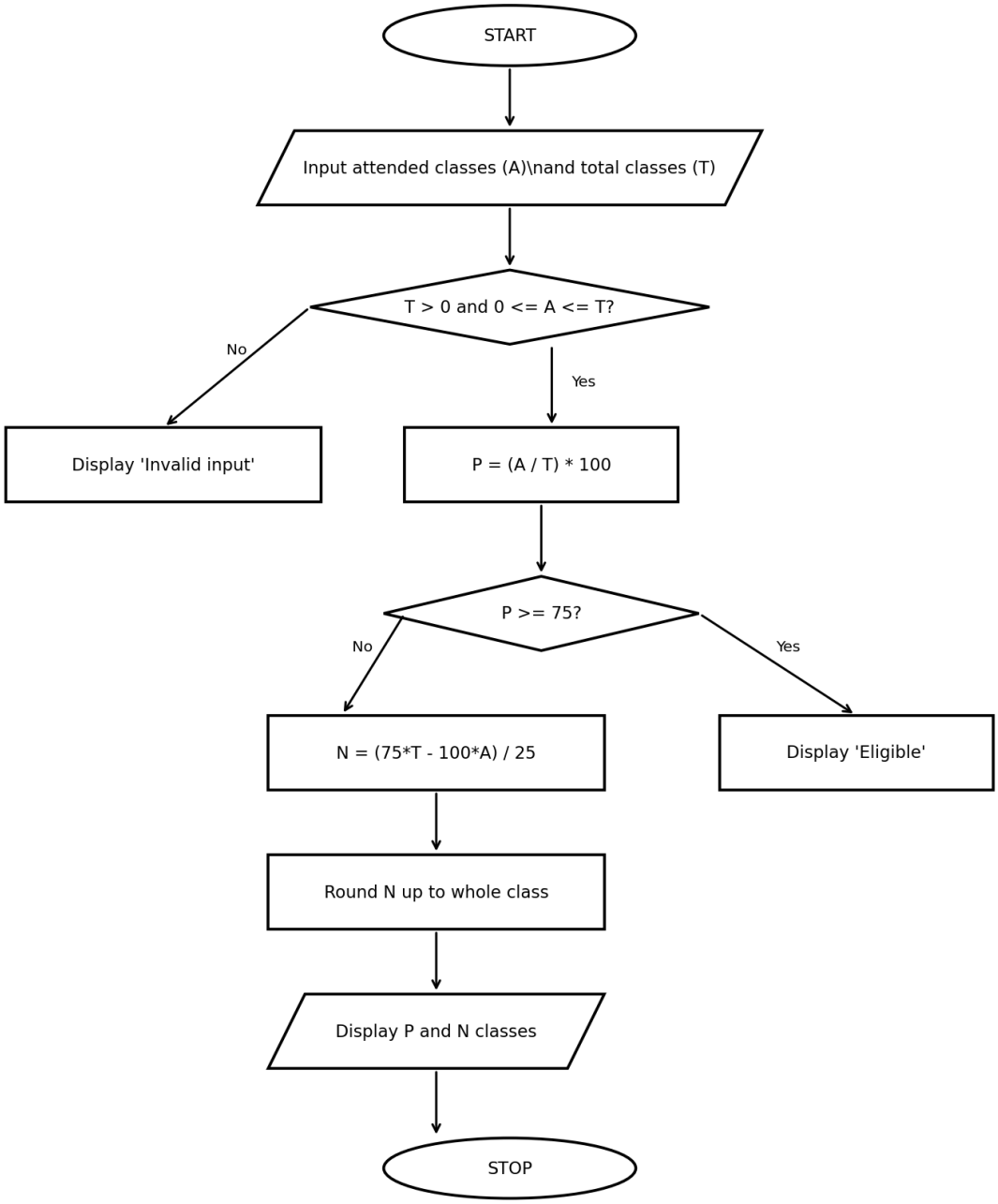
6. Otherwise calculate N = (75 × T – 100 × A) / 25 and round N up to the next whole number.

7. Display the percentage and the number of classes to attend.

8. Stop.

5. Flowchart

The flowchart shows input, validation, percentage checking, and the recovery calculation.



Oval = Start/Stop; parallelogram = Input/Output; diamond = decision; rectangle = calculation/process.

6. Pseudocode

```
START
INPUT A, T
IF T <= 0 OR A < 0 OR A > T THEN
    PRINT "Invalid input"
ELSE
    P = (A / T) * 100
    PRINT P
    IF P >= 75 THEN
        PRINT "Attendance requirement satisfied"
    ELSE
        N = (75 * T - 100 * A) / 25
        ROUND N UP TO THE NEXT WHOLE NUMBER
        PRINT N, "classes to attend"
    END IF
END IF
STOP
```

7. Python Source Code

```
import math

attended = int(input("Enter attended classes: "))
total = int(input("Enter total classes: "))

if total <= 0 or attended < 0 or attended > total:
    print("Invalid input")
else:
    percentage = (attended / total) * 100
    print("Attendance Percentage =", round(percentage, 2), "%")

    if percentage >= 75:
        print("Attendance requirement satisfied")
    else:
        classes = math.ceil((75 * total - 100 * attended) / 25)
        print("Attend", classes, "more classes continuously")
```

Simple explanation: attended means classes attended. total means classes conducted so far. percentage calculates current attendance. The if-else checks whether attendance is at least 75%. math.ceil() rounds up to a whole class because a student cannot attend part of a class.

8. Test Cases

Test	Attended (A)	Total (T)	Expected result
1	75	100	75.00% — requirement satisfied
2	60	100	60.00% — attend 60 more classes
3	70	100	70.00% — attend 20 more classes
4	40	50	80.00% — requirement satisfied
5	0	0	Invalid input

9. Sample Output

```
Enter attended classes: 60
Enter total classes: 100
Attendance Percentage = 60.0 %
Attend 60 more classes continuously
```

Another example:

```
Enter attended classes: 70
Enter total classes: 100
```

```
Attendance Percentage = 70.0 %
Attend 20 more classes continuously
```

10. Conclusion

Through this case study, I learned to calculate attendance percentage and use if-else conditions in Python. I also learned to calculate how many consecutive classes are needed to reach the 75% requirement. The program validates inputs and displays a simple result.